

The key idea of the paper is to present a cellular control plane that provides users an abstraction of reliable access to cellular services while ensuring low latency. As 5G aims to provide ultra low latency and high reliability, new design of cellular core is required, particularly in the control plane. The authors state that the existing cellular cores are deployed in remote data centers, which can cause path inflation. In 5G and beyond, these functions are expected to be moved to the edge, resulting in a highly distributed core network architecture. The main challenge in providing low latency stems from the cellular network core, which needs to maintain dynamic state per user to support mobility.

The authors state that control plane processing of traffic has a direct impact on latency, which will be exacerbated with 5G. The authors present the challenges in Edge based cellular core and present Neutrino which aims to tackle them. The first challenge is User Equipment (UE) core state consistency. Inconsistent core state between the user equipment and the core network can cause significant disruptions in data access on 4G LTE networks, which will amplify with 5G. The authors argue that to tackle this issue, cellular control plane must provide state consistency guarantees. Second challenge is slow state updates between user devices, base stations and cellular control plane, which are a key reason for delays in control traffic. The existing proposals use ASN.1, but it can become a bottleneck for latency sensitive applications. The third challenge is frequent control handovers.

To tackle these issues, the paper presents Neutrino. Neutrino achieves core state consistency by using a consistency protocol that minimizes service disruption under failures by ensuring that the devices are always able to receive Read your Writes consistency. The protocol provides; 1. fast failure recovery and small failure-free overhead, fast in memory message logging and two level failure recovery protocol. The protocol uses primary backup state replication scheme consisting of per procedure check pointing and non blocking synchronization of state. 2. Fast serialization scheme for cellular state updates by optimizing flat buffers, which is 19.1x faster than ASN.1 with small increase in bandwidth usage, and 3. proactive georeplication that can lead to 7x improvement in control plane handovers.

The proposed work offers massive reductions in control procedure completion times. The results show an improvement of 3.1x without failures and up to 5.6x under control plane failures over existing EPC. Neutrino shows 2x improvement in bursty IoT control traffic handling and offers 37x reduction in video startup latency. It also outperforms other recent proposals by up to 3.4x.

The paper demonstrates promising work in control plane optimizations which could significantly improve 5G. Some of these optimizations may also be applicable to existing networks like 4G LTE. However, I am skeptical about some of the claims made in the paper, particularly the one related to achieving 37x reduction in video startup latency. With the global distribution of CDNs of popular video streaming sites like YouTube, the startup latency is fairly low to begin with, even on 3G network. My understanding here is that the results are simulated, and actual gains would be much lower comparatively.